

ECEN-601: Homework 6

Due on November 18, 2025 at 11:59 pm

Instructor: Dr. Jean-François Chamberland

Brody Jordan
431005357

Problem 1

Consider $\mathbb{R}^4$ with the standard inner product. Let W be the subspace of $\mathbb{R}^4$ consisting of all vectors which are orthogonal to both $\alpha = (1, 0, -1, 1)$ and $\beta = (2, 3, -1, 2)$. Find a basis for W .

Solution

Problem 2

Let V be an inner product space, and let α and β be vectors in V . Show that $\alpha = \beta$ if and only if $\langle \alpha | \gamma \rangle = \langle \beta | \gamma \rangle$ for every γ in V .

Solution

Problem 3

Let

$$\mathbf{p}_1 = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix} \quad \mathbf{p}_2 = \begin{bmatrix} 4 \\ -2 \\ -6 \\ -7 \end{bmatrix} \quad \mathbf{p}_3 = \begin{bmatrix} 3 \\ 4 \\ -2 \\ 1 \end{bmatrix}$$

and

$$\mathbf{x} = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 7 \end{bmatrix}.$$

Determine the best approximation, $\hat{\mathbf{x}}$, of the vector $\mathbf{x}$ by vectors in $W = \text{span}\{\mathbf{p}_1, \mathbf{p}_2, \mathbf{p}_3\}$. Also, determine the projection of $\mathbf{x}$ onto the orthogonal complement of W .

Solution

Problem 4

Let S be a subset of an inner product space V . Show that $(S^\perp)^\perp$ contains the subspace spanned by S . When V is finite-dimensional, show that $(S^\perp)^\perp$ is the subspace spanned by S .

Solution

Problem 5

Let V be a finite-dimensional inner product space, and let $\mathcal{B} = (\alpha_1, \dots, \alpha_n)$ be an orthonormal basis for V . Let T be a linear operator on V ; and A , the matrix of T in the ordered basis $\mathcal{B}$. Prove that

$$A_{ij} = \langle T\alpha_j \mid \alpha_i \rangle$$

Solution

Problem 6

Suppose $V = W_1 \oplus W_2$ and that f_1 and f_2 are inner products on W_1 and W_2 , respectively. Show that there is a unique inner product f on V such that

- (a) $W_2 = W_1^\perp$;
- (b) $f(\alpha, \beta) = f_k(\alpha, \beta)$, when α, β are in $W_k, k = 1, 2$.

Solution